BMS-in-the-Loop Traffic Management for Urban Drone Fleets: Chance-Constrained Energy Reserves, Pad Arbitration, and Energy-Weighted ADMM Deconfliction

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***Abstract*— Battery-electric drones are limited by their remaining energy, yet unmanned traffic management (UTM) algorithms assign priority and deconfliction effort by mission class and geometry, and battery health management decides for one vehicle at a time. This paper places the battery management system (BMS) of every drone inside the decision loop of a hierarchical fleet traffic manager. Each drone carries a 1-RC Thevenin pack model with per-cell spread and a lumped thermal model; its BMS estimates the state of charge, polarisation voltage and capacity with an extended Kalman filter from noisy, biased sensors. Every energy decision uses a chance-constrained usable-energy bound together with a weakest-cell voltage-sag check. The drone traffic controller uses the same state to give critical drones right of way, to arbitrate landing pads by remaining endurance, and to set bounded, energy-margin-dependent weights in a consensus-ADMM conflict resolver. The approach is implemented in MASooda, an open simulator built on road-aligned corridors traced from the IIT Hyderabad campus, and evaluated in 460 paired-seed runs with 100 drones. The estimator kept the state-of-charge error below 1.0% and never overestimated usable energy in nominal flight (error below 1.3% in all fault conditions); after a sudden 30% capacity loss it overestimated energy in 9% of samples during a learning transient of about two minutes. In a low-charge fleet, the chance-constrained check kept 6.4 more drones per run on the ground and avoided 3.5 in-flight diversions relative to a BMS with perfect knowledge, at an unchanged number of parcels delivered. Health-aware pad arbitration and ADMM weighting produced no measurable improvement in safety, deliveries or emergency delay in the tested scenarios; in one of 20 battery-fault runs, a drone holding short for a critical drone stood in its path, and the critical drone's failed cell reached its voltage limit during the avoidance manoeuvre. No energy-depletion event occurred in any arm. The evaluation also exposes a limitation: at 44 °C every overheating drone reached a thermal fault before its diversion began, which motivates predictive thermal responses.**

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[†]Code and simulator: `https://github.com/rlv-iith/MASooda`.

Index Terms*— *Unmanned aircraft system traffic management (UTM), battery management system, state-of-charge estimation, chance constraints, ADMM, multi-agent systems.

1 Introduction

Low-altitude delivery and emergency drones are battery-electric, and their endurance is the binding constraint on almost every operational decision: whether to take off, whether to continue, where to land, and how much a conflict manoeuvre may cost. Policy documents already anticipate this. India's National UTM Policy Framework lists real-time health and status monitoring and contingency management among UTM services [1]; NASA's In-time Aviation Safety Management System concept names vehicle-health monitors and battery prognostics as services that can trigger automatic mitigation [2]; the FAA UTM concept of operations, by contrast, leaves endurance to the operator [3].

The traffic-management algorithms themselves are, however, largely energy-agnostic. Tactical deconfliction priorities are derived from the operation class and manoeuvrability [4], separation assurance between heterogeneous fleets models equipage and sensing rather than energy [5], and distributed trajectory optimisation handles collision and kinematic constraints only [7]. When energy enters, it does so before the flight [8], at the arrival schedule [9], or as a cost that is measured after conflicts have been resolved [10]. Conversely, battery prognostics and health management (PHM) for electric aircraft is mature [11–15], but its decisions—abort, replan, re-route to a landing site—are taken by a single vehicle that does not know the surrounding traffic or the occupancy of the landing pads [16–19].

This paper closes the loop between the battery management system (BMS) of each drone and the traffic manager of the fleet. The setting is MASooda, a hierarchical multi-agent traffic-management simulator built on road-aligned corridors traced from the campus of IIT Hyderabad. Each drone carries an equivalent-circuit battery model with per-cell spread and a lumped thermal model, and its BMS estimates the state of charge, the polarisa-

tion voltage and the capacity with an extended Kalman filter (EKF) from noisy, biased sensors. Every energy decision, onboard or at the traffic controller, uses a one-sided chance-constrained energy bound and a weakest-cell voltage check rather than the true state. The traffic controller then uses the same BMS state in three places: it gives drones in a critical health state right of way, it arbitrates landing pads by remaining endurance, and it raises a drone's weight in the ADMM conflict resolver continuously as its energy margin shrinks.

The contributions are:

1. An open, reproducible hierarchical traffic-management simulator for urban drone fleets that combines local car-following and boids-style coordination, consensus-ADMM conflict resolution, a priority-aware drone traffic controller (DTC) with a reserved emergency layer, trusted-range and W-MSR filtering of position reports, and layer-aware signal temporal logic (STL) monitoring (Section 3).

2. A BMS-in-the-loop decision layer: a 1-RC Thevenin pack model with weakest-cell usable energy, an EKF on $(\mathrm{SoC}, V_{RC}, 1/Q)$, and a feasibility test that combines a chance-constrained energy bound with a voltage-sag check (Sections 4–5).

3. Two fleet-level uses of that state: endurance-ordered landing-pad arbitration with hold-short of non-urgent arrivals, and a continuous, bounded ADMM weight derived from the energy margin (Section 6).

4. A paired-seed evaluation on the campus map that separates the contribution of the traffic manager from that of the state estimate, together with architecture, resilience, scaling and link-outage experiments (Section 8).

Section 2 reviews related work, Sections 3–6 describe the method, Section 7 the implementation, Section 8 the experiments, and Section 9 the limitations.

2 Related Work

2.1 Priority-Aware and Distributed Deconfliction

Jover *et al.* [4] extend the PCAN tactical algorithm for U-space with a priority derived from the operation class and manoeuvrability; lower-priority drones absorb most of the deviation (up to 8% more distance and 17% more flight time) with no conflicts among 100 UAVs. Learning-based separation assurance between heterogeneous fleets [5] and edge-AI conflict management [6] optimise separation, latency and accuracy, but do not model the energy or health of the vehicles. ADMM has been applied to robust multi-robot trajectory optimisation with collision and kinematic constraints [7]; MASooda uses consensus ADMM [22] for short-horizon velocity negotiation in the same spirit. None of these works lets the energy state decide who manoeuvres.

2.2 Energy in Traffic Management

Liu and Sengupta plan flights to minimise energy [8]. Kleinbekman *et al.* [9] include each eVTOL's state of charge in a mixed-integer arrival-sequencing model, so that the latest feasible arrival time follows from the remaining charge. KrisshnaKumar *et al.* [20] include battery level in the state and reward of a learned vertiport take-off and landing scheduler. Zongo and Wei [10] quantify the energy overhead of conflict resolution in high-density airspace: the median is below 1.5%, but the tail reaches 44%, and the battery state does not feed back into how conflicts are resolved. SkyGrid [21] balances energy and throughput at aerial intersections.

2.3 Battery Health Management for a Single Vehicle

Battery health management for electric UAVs predicts end of discharge with particle filters [11] or unscented Kalman filters [12], provides flight-tested remaining-flying-time alarms [13, 14], and couples prognostics with thermal models [15]. Decision layers built on these estimates are single-vehicle: real-time mission feasibility checks that re-route to the nearest landing site [16], risk-based trajectory selection [17], health-aware mission and altitude planning with reinforcement learning [18], and onboard contingency managers that move from nominal flight to "land now" [19]. Extended Kalman filtering of equivalent-circuit models is the established BMS technique [23, 24].

2.4 Standards and Regulation

ASTM F3548-21 defines operational-intent states, including non-conforming and contingent, for UTM service-supplier interoperability [25]. India's drone rules limit the green zone to 400 ft above ground level and to 200 ft in the 8–12 km band around an airport [26]; MASooda's layer stack is placed inside the 400 ft ceiling.

2.5 Positioning

Table 1 summarises the comparison. Prior work either manages traffic without the live battery state, or uses the battery state for one vehicle without the traffic. The present work uses an estimated, uncertainty-aware BMS state inside tactical fleet decisions—priority, pad allocation and ADMM weights—and measures the effect of each element separately.

Table 1: Battery state in related traffic-management and health-management work

Work	Scope	Battery information used	Decision it affects	Tactical deconfliction
Jover *et al.* [4]	fleet (U-space)	none	priority by operation class	yes
Zongo and Wei [10]	fleet (en route)	energy overhead, measured	none (reserve sizing)	yes (MVP)
Kleinbekman *et al.* [9]	arrivals at a vertiport	SoC constraint	arrival schedule	no
KrisshnaKumar *et al.* [20]	vertiport	battery level	take-off/landing scheduling	no
Taye *et al.* [16]	single vehicle	predicted voltage	reroute to landing site	no
Quinones-Grueiro *et al.* [17]	single vehicle	SoC, remaining life	trajectory selection	no
Montazeri *et al.* [18]	single vehicle	forecast voltage curve	mission order, altitude	no
This work	**fleet (campus)**	**EKF SoC, V_{RC}, capacity; cell voltages**	**priority, pad, ADMM weight, divert**	**yes (ADMM)**

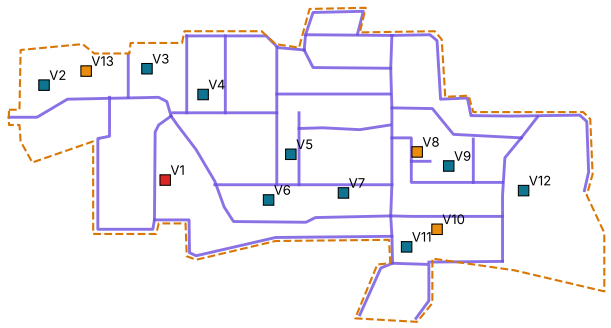


Figure 1: Road-aligned corridor graph of the IIT Hyderabad campus used in all experiments (traced from the campus map). Squares mark vertiports; the dashed line is the geofence.

3 System Overview

3.1 Airspace

Corridors follow the road network: 28 road polylines traced from the IIT Hyderabad campus map form a graph of 105 junctions with 13 vertiports (one hospital, three depots, nine hubs), enclosed by a geofence (Fig. 1). Each corridor is a tube of width 20 m; inside it, each direction keeps to its own lane. Seven altitude layers are stacked 15 m apart starting at 30 m above ground (L_0–L_6 = 30–120 m); following the semicircular rule, eastbound traffic uses even and westbound traffic odd layers, and L_6 is reserved for medical emergencies. The campus is partitioned into six sectors; each (sector, layer) cell has a nominal capacity of 15 drones. Three mission types share the airspace: emergency (priority 0), passenger eVTOL (priority 1) and parcel delivery (priority 2).

3.2 Hierarchical Architecture

Fig. 2 shows the two tiers. Each drone follows its corridor route with a reference speed that brakes to stop at the pad, a car-following law $v \le (g - d_0)/h$ with $d_0 = 12$ m and $h = 1$ s, boids-style separation, alignment and along-road cohesion [27], and closest-point-of-approach avoidance. Drones within 40 m exchange position reports over vehicle-to-vehicle (V2V) links. Before each tick, conflicting groups (pairs closer than the ADMM detection radius of 20 m on the same layer) negotiate their next positions with consensus ADMM (Section 3.3).

The DTC runs every 2 s. It counts the load of every (sector, layer) cell, keeps emergencies on the reserved layer, moves drones out of over-capacity cells subject to a safety gate that defers a layer change if the target layer is occupied nearby, and settles layer requests from drones that are stuck behind slower traffic; drones also negotiate layers among themselves over V2V with consensus-based congestion prices. Operators can hold, reroute, ground, return to base, quarantine, impose speed limits, ground stops and temporary flight restrictions. If the DTC link fails, drones continue on their last directive and commands are queued.

Position reports pass a trusted-range check: a report that disagrees with the sensed range by more than 3 m is moved along its bearing to the sensed range, and W-MSR [28] with $f = 1$ fuses neighbour reports. Safety is monitored with the STL formula $\varphi = \mathbf{G}(d_{ij}(t) \ge d_{\min})$, $d_{\min} = 8$ m, through its robustness $\rho(t) = \min_{i \ne j} d_{ij}(t) - d_{\min}$ [29, 30], where d_{ij} is the layer-aware distance $\|(\Delta x, \Delta y, \Delta L \cdot 15\,\mathrm{m})\|$.

3.3 Consensus ADMM Conflict Resolution

For a conflicting group $\mathcal{G}$ with positions p_i and intended positions $t_i = p_i + v_i\,\Delta t$, the resolver solves

$$\begin{aligned} &\min_{\{x_i\}} \sum_{i \in \mathcal{G}} \tfrac{w_i}{2}\,\|x_i - t_i\|^2 \\ &\text{s.t. } n_{ij}^{\top}(x_i - x_j) \ge d_s \quad \forall (i,j) \in \mathcal{P}, \\ &\|x_i - p_i\| \le v_{\max}\Delta t \quad \forall i \in \mathcal{G}, \end{aligned} \tag{1}$$

where n_{ij} is the unit vector from p_j to p_i (a convexification of the separation constraint), $d_s = 1.01\,d_{\min}$, $\mathcal{P}$ contains the pairs on the same layer that can reach each other within one tick, $v_{\max} = 15$ m/s and $\Delta t = 0.1$ s. Each

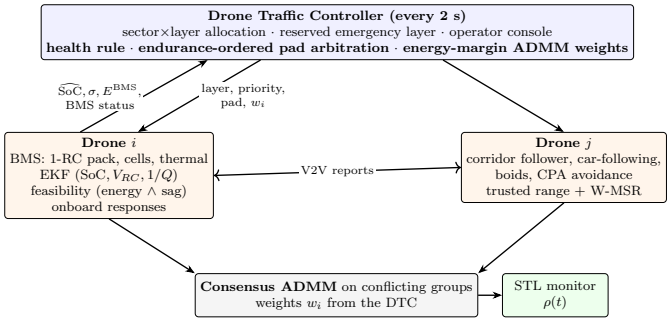


Figure 2: MASooda architecture. Bold items are the BMS-in-the-loop elements introduced in this paper: the BMS estimate and feasibility test onboard, and their three uses by the DTC.

agent keeps one copy of its variable per constraint; the x-update is closed form,

$$x_i = \frac{w_i t_i + \rho \sum_k (z_{ik} - u_{ik})}{w_i + \rho K_i}, \tag{2}$$

with K_i the number of constraints that involve i, followed by a projection of each copy onto its constraint set and a scaled dual update [22]. The parameters are $\rho = 1$, absolute and relative tolerances 10^{-3} and 10^{-4}, and at most 50 iterations; a group that does not converge keeps its intended velocities and is counted as a fallback. In the baseline, w_i is the class weight $w^c = 10, 3, 1$ for priorities 0, 1, 2.

4 Vehicle Energy and Battery Model

4.1 Power

With thrust $T = mg$, hover-induced velocity $v_h = \sqrt{T/(2\rho_a A)}$ and airspeed $v = \|v_g - v_w\|$ (ground velocity minus wind), the mechanical and electrical power follow momentum theory with parasitic drag [31]:

$$v_i = \sqrt{-v^2/2 + \sqrt{v^4/4 + v_h^4}}, \tag{3}$$

$$P_{\text{mech}} = T v_i/\text{FM} + \tfrac{1}{2}\rho_a C_D A_f\, v^3 \ (+ T v_{\text{climb}} \text{ in a climb}), \tag{4}$$

$$P_{\text{el}} = P_{\text{mech}}/(\eta\, h_m) + P_{\text{av}}, \tag{5}$$

where FM is the figure of merit, η the propulsive efficiency, $h_m \in (0, 1]$ the motor health and P_{av} the avionics load. Air density ρ_a follows the standard atmosphere at the site elevation (542 m) and the ambient temperature; at 35 °C it is 1.074 kg/m^3, which raises hover power by 3.4% relative to 15 °C. Table 2 lists the parameters of the three vehicle types.

Table 2: Vehicle and pack parameters

	Delivery	Emergency	eVTOL
Mass + payload (kg)	6 + 2	10 + 3	450 + 100
Rotors × diameter (m)	4 × 0.38	6 × 0.55	16 × 1.5
FM / η	0.65 / 0.80	0.65 / 0.80	0.70 / 0.85
$C_D A_f$ (m^2)	0.08	0.12	1.2
Avionics (W)	25	40	500
Cells in series / V_{nom} (V)	6 / 22.2	12 / 44.4	108 / 400
Capacity (Wh)	500	800	25 000
R_0 (Ω) / $C_{\max}$	0.02 / 5	0.04 / 5	0.05 / 5
Pack heat capacity (J/K)	2500	4000	139 000

4.2 Equivalent Circuit and Cells

The pack is a 1-RC Thevenin circuit [24]. The n_s series cells share one current but have seeded SoC offsets $\delta_k \sim \mathcal{N}(0, 0.4\%)$ and resistance factors $\varepsilon_k \sim \mathcal{N}(0, 3\%)$. Because the cell open-circuit-voltage (OCV) curve is piecewise linear, the pack OCV is $\text{OCV} = n_s\, \text{OCV}_c(\text{SoC} + \bar{\delta})$. With R_0 scaled for ageing and cold, $R_1 = 0.5R_0$ and $\tau = R_1 C_1 = 20$ s,

$$I = \frac{E - \sqrt{E^2 - 4R_0 P_{\text{el}}}}{2R_0}, \qquad E = \text{OCV} - V_{RC}, \tag{6}$$

$$V = \text{OCV} - V_{RC} - I R_0, \tag{7}$$

$$V_{RC}^{+} = V_{RC}\, e^{-\Delta t/\tau} + R_1(1 - e^{-\Delta t/\tau}) I, \tag{8}$$

$$\text{SoC}^{+} = \text{SoC} - (I + I_f)\, \Delta t/(3600\, Q), \tag{9}$$

where $Q = Q_{\text{nom}}\, \text{SoH}$ and I_f is an internal fault current (below). If $E^2 < 4R_0 P_{\text{el}}$, the requested power cannot be delivered. The energy drawn from the cells per tick, $\text{OCV}\, I\, \Delta t$, equals the delivered energy plus the heat $I^2 R_0 + I V_{RC}$; the test suite checks this balance to 10^{-9}.

Weakest-cell usable energy. Series cells give up the same charge, so the pack stops when its weakest cell reaches $\text{SoC}_{\min} = 5\%$. With $s_k = \text{SoC} + \delta_k$ and span $\Delta = \min_k s_k - \text{SoC}_{\min}$,

$$E^{\text{use}}(\text{SoC}, Q) = Q \sum_{k=1}^{n_s} \int_{s_k - \Delta}^{s_k} \text{OCV}_c(s)\, ds. \tag{10}$$

For identical cells this reduces to the usual $Q\, n_s \int_{\text{SoC}_{\min}}^{\text{SoC}} \text{OCV}_c$; after a cell failure the whole string loses the failed cell's missing span.

4.3 Temperatures, BMS States and Faults

Pack and motor temperatures follow first-order lumped models [32], $C_p \dot{T}_p = I^2 R_0 + I V_{RC} + q_f - (h_0 + h_1\sqrt{v})(T_p - T_{\text{amb}})$ and $C_m \dot{T}_m = (1 - \eta h_m) P_{\text{el}}^{\text{prop}} - (h_{m0} + h_{m1}\sqrt{v})(T_m - T_{\text{amb}})$, so a hovering drone cools worse than a cruising one. Every 0.5 s the BMS computes the cell voltages $V_k = \text{OCV}_c(s_k) - I R_c(1 + \varepsilon_k)$, the imbalance $\max_k V_k - \min_k V_k$ and five categories (charge, pack temperature, motor temperature, imbalance, current), each with WARNING, CRITICAL and FAULT thresholds and hysteresis (Table 3). The drone's status is the worst category. Five

Table 3: BMS thresholds (hysteresis: 2% SoC, 3 °C, 10 mV)

Category	WARNING	CRITICAL	FAULT
SoC (estimated)	$< 30\%$	$< 15\%$	any cell < 3.0 V
Pack temperature	> 55 °C	> 65 °C	> 75 °C
Motor temperature	> 90 °C	> 110 °C	—
Cell imbalance	> 50 mV	> 100 mV	—
Current	$> 0.8\, I_{\max}$	$> I_{\max}$	infeasible
Sensor	—	—	injected fault

faults can be scheduled: capacity loss ($Q \leftarrow 0.7Q$), cell failure (one cell -15% SoC and $3\times$ resistance), overheating (an internal dissipation $q_f = 0.6\, P_{\text{hover}}$ that also drains charge as $I_f = q_f/\text{OCV}$), motor degradation ($h_m = 0.8$) and GNSS degradation (a sensor fault).

5 BMS State Estimation and Chance-Constrained Feasibility

5.1 Extended Kalman Filter

The BMS does not observe the true state. It measures the pack current with a seeded gain error ($\sigma = 0.5\%$) plus noise of 0.2 A and 1% of the reading, and the pack voltage with noise $5\sqrt{n_s}$ mV. An internal fault current is not seen by the current sensor. The estimator state is $x = (\text{SoC}, V_{RC}, q)$ with $q = 1/Q$, so that a capacity change is learnt rather than known. With measured current $\tilde{I}$,

$$\widehat{\text{SoC}}^{-} = \widehat{\text{SoC}} - \tilde{I}\,\Delta t\,\hat{q}/3600, \tag{11}$$

$$\hat{V}_{RC}^{-} = \hat{V}_{RC}\, e^{-\Delta t/\tau} + R_1(1 - e^{-\Delta t/\tau})\tilde{I}, \tag{12}$$

$$\hat{V} = n_s\, \text{OCV}_c(\widehat{\text{SoC}}^{-} + \bar{\delta}) - \hat{V}_{RC}^{-} - R_0\tilde{I}, \tag{13}$$

and the measurement Jacobian is $H = [\, n_s\, \text{OCV}_c'(\widehat{\text{SoC}}^{-} + \bar{\delta}),\ -1,\ 0\,]$. The cell offsets $\bar{\delta}$ and R_0 are assumed to be identified online from the cell voltages. Process noise is 10^{-6} s^{-1} on SoC, 10^{-4} V^2/s on V_{RC} and $(5 \cdot 10^{-4}\,\text{s}^{-1})\,\hat{q}^2$ on q; the filter starts from the rest-voltage SoC with $\sigma = 1.5\%$ and a capacity uncertainty of 5%. Sensor noise is generated from a counter-based hash of (seed, drone, sample index), so a drone's measurements do not depend on which other drones are flying, and runs replay exactly.

5.2 Chance-Constrained Energy

Every decision uses the lower one-sided bound

$$E^{\text{BMS}} = E^{\text{use}}\left(\widehat{\text{SoC}} - z\,\sigma_{\text{SoC}},\ \frac{1}{\hat{q} + z\,\sigma_q}\right), \qquad z = 1.645, \tag{14}$$

which, for a Gaussian estimation error, overestimates the usable energy with probability of about 5% per factor. The BMS charge levels and the SoC reported to the DTC are the estimate; the true SoC only drives the physics and the *energy-depleted* counter.

5.3 Feasibility Test

The energy needed to fly a route of length L at cruise speed v_c and land is

$$E^{\text{need}}(L) = P_{\text{el}}(v_c)\,\frac{L}{v_c}\,(1 + \mu) + P_{\text{el}}(0)\,t_{\text{land}}, \tag{15}$$

with reserve margin $\mu = 0.2$ and $t_{\text{land}} = 30$ s. A route is *feasible* if $E^{\text{need}}(L) \le E^{\text{BMS}}$ and the weakest cell stays above $V_{\text{cut}} + \delta_V = 3.05$ V under the landing-hover load at the SoC left on arrival; before take-off it must also carry a climb at the current SoC. The voltage condition catches a high-resistance cell that would reach cut-off long before its charge runs out. Before take-off the drone must also be able to lift its mass: the hover-plus-climb current must not exceed $C_{\max} Q_{\text{nom}}$.

6 Health-Aware Traffic Management

6.1 Onboard Responses

Every second, each airborne drone checks its status and the feasibility of the rest of its route. A FAULT lands it in place. A CRITICAL status diverts it to the nearest feasible pad; an overheating-only CRITICAL first limits the speed to 60% of cruise and diverts after 30 s. An infeasible remaining route triggers a low-reserve diversion. A drone whose weakest cell is within three sag margins of cut-off flies at its minimum-power airspeed. A failed preflight check is repeated every 30 s rather than every tick, so that a noisy estimate cannot be retried until it happens to pass.

6.2 DTC Health Rule and Pad Arbitration

A cooperative drone at BMS CRITICAL or FAULT receives priority 0 for right of way, keeps its layer (a climb would cost energy) and becomes a landing candidate for its pad, as do all diverting drones. Each pad is given to the candidate with the least endurance

$$\tau_i = \frac{3600\, E_i^{\text{BMS}}}{P_{\text{el},i}(0)}, \tag{16}$$

replaced by $\min(\tau_i, \tau_c)$ for a drone at CRITICAL or FAULT, with $\tau_c = 120$ s, because a critical pack may fail at any time. Algorithm 1 gives the procedure. While a reservation is active (its holder is within 150 m of route distance of the pad), departures from the pad are held and other arrivals stop 60 m short and loiter at the edge of the tube, so the urgent drone can pass; a medical emergency holds short only for a CRITICAL or FAULT landing. Diversions prefer pads that are not reserved for another drone and, unless the diverting drone is a medical emergency, pads other than the hospital.

Algorithm 1 Endurance-ordered pad arbitration (every 1 s)

Require: airborne drones $\mathcal{A}$, previous reservations R, swap threshold $\Delta\tau = 15$ s

$\mathcal{C} \leftarrow \{i \in \mathcal{A} : \text{status}_i \geq \text{CRITICAL} \vee i \text{ diverting}\}$
for all pads p with candidates $\mathcal{C}_p \subseteq \mathcal{C}$ **do**
  $i^\star \leftarrow \arg\min_{i \in \mathcal{C}_p} \tau_i$
  if $R[p] \in \mathcal{C}_p$ **and** $\tau_{R[p]} - \tau_{i^\star} < \Delta\tau$ **then**
    $i^\star \leftarrow R[p]$ {no flip-flopping}
  end if
  $R'[p] \leftarrow i^\star$
end for
return R' {active once $L^{\text{rem}}_{R'[p]} \leq 150$ m}

6.3 Energy-Margin ADMM Weights

The DTC also computes, every second, the energy margin $m_i = E_i^{\text{BMS}}/E_i^{\text{need}}(L_i^{\text{rem}})$ of every cooperative drone and its weight

$$w_i = w_i^c + (w^{\text{top}} - w_i^c)\, u_i, \qquad u_i = \text{clip}\left(\frac{m_{\text{ref}} - m_i}{m_{\text{ref}} - 1}, 0, 1\right), \tag{17}$$

with $u_i = 1$ at CRITICAL or FAULT, $m_{\text{ref}} = 2$ and $w^{\text{top}} = 10$, the emergency class weight. These weights replace the class weights in (1).

Proposition 1. *(i) $w_i^c \leq w_i \leq w^{\text{top}}$; (ii) w_i is non-increasing in m_i and equals w_i^c for $m_i \geq m_{\text{ref}}$; (iii) the feasible set of* (1) *does not depend on $\{w_i\}$; (iv) in the x-update* (2)*, the weight given to agent i's own target, $w_i/(w_i + \rho K_i)$, is increasing in w_i.*

Proof. (i) and (ii) follow from $u_i \in [0, 1]$, which is non-increasing in m_i and zero for $m_i \geq m_{\text{ref}}$, and from $w^{\text{top}} \geq w_i^c$. (iii) The weights appear only in the objective. (iv) $\partial/\partial w\,[w/(w + c)] = c/(w + c)^2 > 0$ for $c = \rho K_i > 0$. □

Hence a healthy drone never outweighs a medical emergency, healthy traffic is solved exactly as before, and the separation constraints are untouched: the weights only decide which feasible point is chosen, shifting the manoeuvre towards drones with more energy margin. They do not guarantee separation; ADMM is a mitigation layer and the separation outcome is measured.

7 Implementation

MASooda is written in Python 3.10+ with NumPy and PyYAML; the live demonstrator streams frames over a standard-library HTTP server to a browser front end written in plain JavaScript and Canvas 2D. Every parameter is read from one configuration file. The health model is vectorised over drones; the EKF runs as batched 3×3 operations. The simulation advances at $\Delta t = 0.1$ s; the BMS runs every 0.5 s, onboard responses and DTC health decisions every 1 s, and the DTC allocation every 2 s. Runs are deterministic for a given seed. The repository contains about 420 automated tests, including energy-balance, estimator-calibration, pad-arbitration and replay-determinism tests. The experiments in Section 8 use the same code path as the live demonstrator.

8 Experimental Evaluation

8.1 Setup

All experiments use the campus map of Fig. 1, a random fleet of 100 drones (10% emergency, 30% eVTOL, 60% delivery) departing within 90 s, and a horizon of 240 s. Four scenarios are used. *Nominal*: random-fleet SoC uniform in 35–100% and SoH in 80–100%, 35 °C ambient. *Low-SoC*: as nominal but SoC uniform in 15–50%. *Battery faults*: nominal plus four delivery drones on long routes with a cell failure (two drones) or a 30% capacity loss (two drones) injected 45 s after departure. *Heat wave*: 44 °C, 2 m/s wind and four delivery drones with an overheating fault 45 s after departure. The fault drones' routes, departure times and initial SoC are drawn from the seed.

Three traffic-manager arms are compared. *Blind*: drones keep all onboard BMS responses (Section 6-A), but the DTC ignores health and ADMM uses class weights. *Rule*: adds the DTC health rule and pad arbitration. *Full*: adds the energy-margin ADMM weights (17). Three knowledge arms are compared under the full manager: an *oracle* BMS that reads the true state, the EKF with $z = 0$, and the EKF with $z = 1.645$ (the default). Further suites compare the four architectures in nominal traffic (hierarchical, centralized, decentralized with ADMM and report defence, decentralized undefended), a position-spoofing attack by two delivery drones from 60 s, fleet sizes of 50–200, and a DTC link outage from 80 to 150 s.

Each cell has 20 seeds (10 for the scaling and outage suites); arms with the same seed share all initial conditions, so differences are computed per seed. Values are means with 95% t-intervals across seeds. In total 460 runs were executed on one platform (Python 3.13, NumPy 2.5); 132 of them were also run on a second machine (Python 3.10, NumPy 2.2). The 160 runs that exercise pad arbitration in the low-SoC and fault scenarios (Rule and Full arms and the knowledge suite) were re-executed on the first platform only, after a fix to the hold-short logic, and are excluded from that comparison. In the 132 common runs every discrete outcome (arrivals, deliveries, diversions, groundings, forced landings, near misses, collisions) was identical and continuous quantities differed only in floating-point detail. Metrics: *energy depleted* (true SoC reaches zero in flight), *forced landings* (landed in place, off any pad), *diversions*, *grounded* (failed pre-flight check), *held short* (arrivals that waited for a reserved pad), *arrived* (fraction of flights that landed at any pad, including diversions), *delivered* (landed at the planned destination), emergency *delay* (travel time from scheduled departure minus free-flow time) and *on time* (delay $\leq \max(10\,\text{s}, 0.2\times$ nominal)), *near misses* (loss-of-

Table 4: EKF validation (40 drones × 3 seeds per condition, samples after the fault)

	Nominal	Cap. loss	Cell fail.	Overheat
SoC RMSE (%)	0.21	0.26	0.28	0.33
max |error| (%)	0.91	0.97	1.24	1.12
mean σ (%)	0.55	0.61	0.59	0.63
within 3σ (%)	100.0	100.0	100.0	100.0
$P(E^{\mathrm{BMS}} > E^{\mathrm{true}})$ (%)	0.0	9.0	0.0	0.0
max $E^{\mathrm{BMS}} - E^{\mathrm{true}}$ (Wh)	-34	37	-43	-78
$\hat{Q}/Q$ at 360 s	1.01	1.01	0.96	0.61

separation episodes, $d_{ij} < 8$ m) and minimum layer-aware separation.

8.2 State Estimation

The estimator was first validated in isolation: 40 delivery drones per condition and seed (3 seeds), starting at 85% SoC, alternating 60 s of cruise at 11 m/s with 20 s of hover for 360 s, with a fault at 60 s. Table 4 and Fig. 3 summarise the results. The SoC error stayed within $\pm 3\sigma$ in all samples and below 1.3% in all conditions. The reported σ is conservative: the mean σ is 1.9–2.6 times the RMS error. In nominal flight, the chance-constrained energy (14) never exceeded the true usable energy; it was always at least 34 Wh (about 7% of the 500 Wh pack) below it, which is the price of the bound. After a sudden 30% capacity loss, the BMS still believes the old capacity until the voltage residual corrects $\hat{q}$: the energy was overestimated in 9% of post-fault samples, by up to 37 Wh, and never after 180 s. A failed cell is visible in the cell voltages and the estimate stays conservative. The hidden drain of an overheating fault is absorbed by the capacity state, which falls to 0.61 of the true capacity, so the energy bound becomes more conservative rather than optimistic.

8.3 Traffic-Manager Ablation

Table 5 reports the manager arms. In nominal traffic no drone reached CRITICAL or diverted, so Rule produced runs identical to Blind in 20 of 20 seeds; the energy-margin weights changed the trajectory in 11 seeds, without a measurable change in any metric. This is the behaviour Proposition 1(ii) predicts: the health layer is inactive when margins are large.

In the low-SoC fleet the arbitration was active: 7.7 ± 2.2 arrivals per run held short of a reserved pad (Full minus Blind, paired). Diversions (0.05 ± 0.32), pre-flight groundings and the fraction of flights that arrived (-0.4 ± 0.4 percentage points) did not change measurably. Emergency delay rose by 0.9 ± 0.8 s (the interval excludes zero: arbitration cost emergency flights a small delay) and the on-time rate of emergency flights changed by -3.3 ± 5.0 points (the interval includes zero). With battery faults, the two cell-failure drones reached CRITICAL on cell imbalance within 0.4 s of the fault and diverted in all 120 cases; the two capacity-loss drones never reached a diversion threshold and continued to their destinations (0 of 120 diverted). Diverted drones landed 34.5 s after the fault on average. Arbitration made 2.5 ± 1.0 arrivals per run hold short and changed the fault-to-landing time by -0.8 ± 2.7 s. In one seed (13), Rule and Full produced a forced landing that Blind did not: the cell-failure drone diverting to its reserved pad met the drone holding short for it at the tube edge, and during the 7-s ADMM manoeuvre that followed, its failed cell fell below 3.0 V under load. The BMS declared FAULT and the drone landed in place at 106 s, 52 s after the fault with 50% SoC; with the same seed in the blind arm it reached the pad at 109.5 s. Holding traffic at the tube edge can thus put it in the path of the drone it is meant to protect, and a failed cell has little voltage headroom for an avoidance manoeuvre. No arm produced an energy-depletion event or a near miss in any scenario.

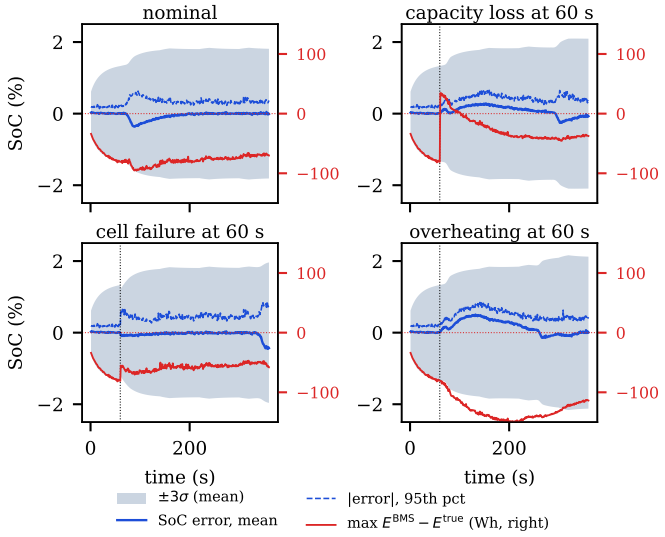


Figure 3: EKF state-of-charge error across 40 drones (seed 1): mean error, 95th percentile of |error| and mean $\pm 3\sigma$ band (left axis), and the largest gap between the chance-constrained and the true usable energy (right axis). Faults at 60 s.

The heat-wave scenario shows a limitation of the onboard thermal response rather than of the traffic manager. In all arms and seeds, all four overheating drones reached a pack-temperature FAULT and landed in place (4.0 of 4 per run). Over the 240 overheating drones, the pack reached 65 °C (CRITICAL) 59 s after the fault on average (56–61 s) and 75 °C (FAULT) 32 s later (30.5–34 s). The onboard response diverts only after 30 s at CRITICAL, so the FAULT arrived 1–5 s into the diversion, long before any pad could be reached.

8.4 What the BMS Knows

Table 6 and Fig. 4 compare the knowledge arms. With battery faults, all three arms produced the same discrete outcomes in every seed, including the forced landing of

Table 5: Traffic-manager ablation (100 drones, 20 paired seeds; per-run mean with 95% CI half-width in parentheses). No energy-depletion event and no near miss occurred in any run.

Arm	Forced	Diversions	Grounded	Held short	Arrived (%)	EMG delay (s)	On time (%)	Min sep. (m)
Nominal								
Health-blind	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	95.7 (0.9)	6.3 (0.8)	97.0 (3.4)	12.5 (0.7)
Rule	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	95.7 (0.9)	6.3 (0.8)	97.0 (3.4)	12.5 (0.7)
Full	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	95.7 (0.9)	6.3 (0.8)	97.0 (3.4)	12.5 (0.7)
Low-SoC fleet								
Health-blind	0.0 (0.0)	5.9 (1.2)	12.1 (1.6)	0.0 (0.0)	88.0 (1.3)	6.2 (0.8)	96.8 (3.8)	12.9 (0.7)
Rule	0.0 (0.0)	5.9 (1.3)	12.1 (1.6)	7.3 (2.1)	87.7 (1.2)	7.1 (1.3)	93.5 (5.9)	12.6 (0.5)
Full	0.0 (0.0)	6.0 (1.4)	12.1 (1.6)	7.7 (2.2)	87.6 (1.3)	7.1 (1.3)	93.5 (5.9)	12.6 (0.5)
Battery faults								
Health-blind	0.0 (0.0)	2.0 (0.0)	0.0 (0.0)	0.0 (0.0)	95.3 (1.1)	6.3 (0.8)	97.0 (3.4)	12.6 (0.4)
Rule	0.1 (0.1)	2.0 (0.0)	0.0 (0.0)	2.5 (1.0)	95.4 (1.1)	6.4 (0.8)	96.5 (3.5)	12.5 (0.4)
Full	0.1 (0.1)	2.0 (0.0)	0.0 (0.0)	2.5 (1.0)	95.4 (1.1)	6.4 (0.8)	96.5 (3.5)	12.5 (0.4)
Heat wave 44 °C								
Health-blind	4.0 (0.0)	4.0 (0.1)	0.0 (0.0)	0.0 (0.0)	91.8 (0.9)	6.3 (0.8)	97.0 (3.4)	12.6 (0.4)
Rule	4.0 (0.0)	3.9 (0.1)	0.0 (0.0)	0.0 (0.0)	91.8 (0.9)	6.3 (0.8)	97.0 (3.4)	12.6 (0.4)
Full	4.0 (0.0)	3.9 (0.1)	0.0 (0.0)	0.0 (0.0)	91.8 (0.9)	6.3 (0.8)	97.0 (3.4)	12.6 (0.4)

Table 6: BMS knowledge under the full manager (20 paired seeds; mean with 95% CI half-width). No depletion occurred; the forced landing with battery faults (seed 13, all arms) is discussed in the text.

	Divert.	Ground.	Arrived (%)	Delivered	Delay (s)
Low-SoC fleet					
Oracle	9.4 (1.2)	5.7 (1.4)	90.8 (1.1)	81.8 (1.7)	7.3 (1.3)
EKF, $z = 0$	9.0 (1.2)	6.5 (1.4)	90.8 (0.9)	82.2 (1.4)	7.2 (1.3)
EKF, $z = 1.645$	6.0 (1.4)	12.1 (1.6)	87.6 (1.3)	81.7 (1.7)	7.1 (1.3)
Battery faults					
Oracle	2.0 (0.0)	0.0 (0.0)	95.4 (1.1)	97.2 (1.1)	6.4 (0.8)
EKF, $z = 0$	2.0 (0.0)	0.0 (0.0)	95.4 (1.1)	97.2 (1.1)	6.4 (0.8)
EKF, $z = 1.645$	2.0 (0.0)	0.0 (0.0)	95.4 (1.1)	97.2 (1.1)	6.4 (0.8)

seed 13: the cell-failure drones diverted on cell imbalance, which does not depend on the SoC estimate, and the capacity-loss drones never came close enough to a threshold for the estimate to change a decision. In the low-SoC fleet, the chance constraint changed when drones were stopped: relative to the oracle it kept 6.4 ± 1.0 more drones per run on the ground and avoided 3.5 ± 1.1 in-flight diversions, while the number of parcels delivered changed by only -0.2 ± 0.7. The fraction of flights that landed anywhere fell by 3.2 ± 0.7 points, because a grounded drone does not count as an arrival while a diverted one does. The EKF without the margin ($z = 0$) stayed close to the oracle. No forced landing occurred in the low-SoC fleet in any knowledge arm.

8.5 Architectures, Resilience and Scaling

Table 7 compares the architectures in nominal traffic. The hierarchical and both decentralized variants delivered the same 95.7% of flights with no near miss and a minimum separation of 12.5 m; their emergency delays were within 0.1 s of each other, so in this scenario the DTC's layer allocation did not change the outcome measurably. The centralized baseline, in which the DTC computes every velocity from reports refreshed at its update rate, had 9.8 ± 1.7 near misses and 1.10 ± 0.45 collisions (pairs closer than 2 m) per run, a minimum separation of 1.7 ± 0.3 m and 88.0% arrivals.

Under the position-spoofing attack (Table 8), the trusted-range check and W-MSR filtering accepted no spoofed report, while the undefended variant accepted 193 ± 109 per run. Separation was unaffected in both cases (near misses 0; minimum separation changed by -0.07 ± 0.10 m): with two spoofers and a 25 m offset, the false reports were not decisive for collision avoidance in this traffic. The DTC outage dropped 35 directives per run but changed no outcome relative to the same seeds without outage, consistent with the small role of the DTC in nominal traffic noted above.

Fig. 5 shows the scaling from 50 to 200 drones. ADMM groups per run grew from 146 to 1200, but groups stayed small: mean iterations remained about 8.7, every group converged, and the 95th-percentile solve time stayed near 0.45 ms. The whole simulation ran at 22.2× real time with 50 drones and 5.7× with 200 on one CPU core. Arrivals fell from 95.8% to 90.9% within the 240 s horizon as pads and junctions congested, with no near miss.

8.6 Case Study: Battery Failure Mid-Route

The scripted battery-failure scenario (100 random drones and two scripted delivery drones) illustrates the mechanisms. At 70 s, cargo drone D-100 loses a cell; its BMS goes CRITICAL on cell imbalance, it diverts to the nearest feasible pad and the DTC reserves that pad. One arrival bound for the same pad holds short at the tube edge, and D-100 lands at 117.9 s, first on its pad. At 80 s, D-101 (31% SoC) loses 30% of its capacity. Its BMS does not know this; it learns the loss from the voltage and declares a low-reserve diversion at 144 s to a different pad, where it lands at 172.7 s. The run ends with no forced landing, no depletion and no near miss. On the

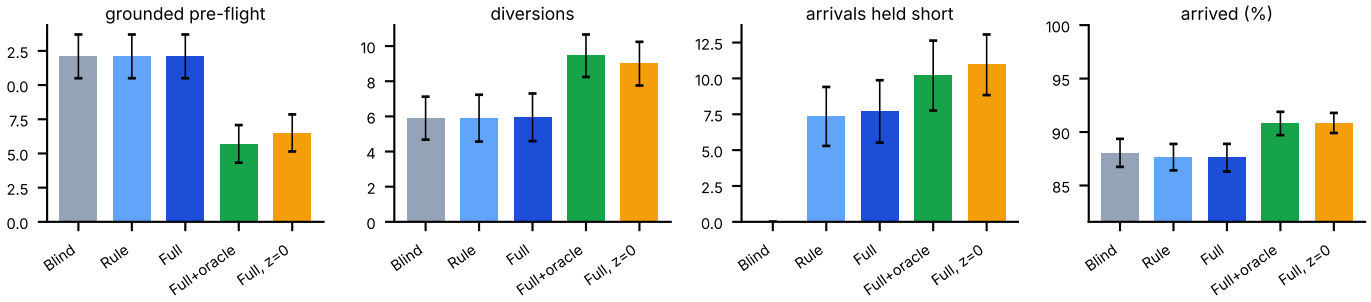


Figure 4: Low-SoC fleet: drones grounded by the pre-flight check, in-flight diversions, arrivals that held short of a reserved pad, and flights that arrived at any pad, per run (mean and 95% CI over 20 paired seeds). Blind, Rule and Full use the EKF with $z = 1.645$; the last two bars use the full manager with an oracle BMS and with the EKF at $z = 0$.

Table 7: Architectures in nominal traffic (100 drones, 20 seeds; mean with 95% CI half-width; RTF: real-time factor on one CPU core)

Architecture	Arrived (%)	Near misses	Viol. ticks	Min sep. (m)	Collisions	EMG delay (s)	ADMM groups	RTF
Hierarchical	95.7 (0.9)	0.00 (0.00)	0.0 (0.0)	12.5 (0.7)	0.00 (0.00)	6.3 (0.8)	432 (74)	11.9 (0.3)
Centralized	88.0 (1.4)	9.85 (1.71)	77.6 (12.4)	1.7 (0.3)	1.10 (0.45)	5.8 (0.4)	0 (0)	12.5 (0.5)
Decentralized	95.7 (0.9)	0.00 (0.00)	0.0 (0.0)	12.5 (0.7)	0.00 (0.00)	6.3 (0.8)	452 (72)	12.4 (0.3)
Decentr., undefended	95.7 (0.9)	0.00 (0.00)	0.0 (0.0)	12.5 (0.7)	0.00 (0.00)	6.3 (0.8)	449 (71)	12.8 (0.4)

Table 8: Spoofing attack (20 seeds) and DTC outage (10 seeds), hierarchical architecture with the full manager. No near miss, separation violation or collision occurred in any of these runs.

	Spoofed acc.	Min sep. (m)	Arrived (%)
No attack	0 (0)	12.5 (0.7)	95.7 (0.9)
Attack, defended	0 (0)	12.5 (0.7)	95.8 (0.9)
Attack, undefended	193 (109)	12.4 (0.7)	95.8 (0.9)
DTC outage	0 (0)	11.9 (1.3)	96.0 (1.0)

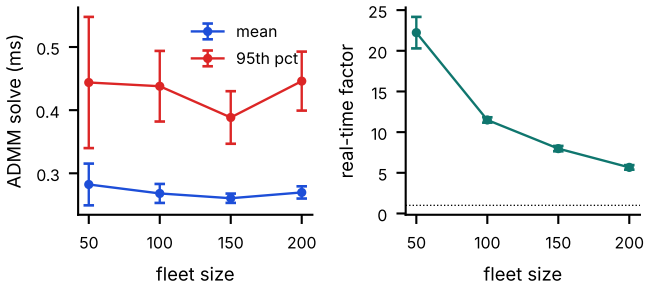


Figure 5: Scaling with fleet size (10 seeds): ADMM solve time per conflict group (left) and real-time factor of the whole simulation on one CPU core (right).

publicly deployed version that preceded the changes described here, the same scenario let three drones land on one reserved pad within 7 s, sent the low-reserve drone to the pad already reserved for the critical one, and flagged the diverting critical drone as having "made no progress".

9 Discussion and Limitations

What the evidence supports. The estimator and the chance-constrained feasibility test behave as designed: the SoC estimate is calibrated (conservatively), the energy bound does not overestimate in nominal flight, and its failure mode—a transient after a sudden, unannounced capacity loss—is bounded in time and measured. At fleet level, the bound moved the decision about marginal drones from the air to the ground: fewer diversions and more pre-flight holds, at an unchanged number of deliveries. The health-aware DTC and ADMM weights did what they specify (pads went to the drone with least endurance, others held short, healthy traffic was untouched), but in these scenarios this produced no measurable gain in safety, deliveries or emergency delay over a BMS-blind traffic manager whose drones already respond onboard, and in one seed it caused a forced landing that the blind manager avoided. The claim of this paper is therefore the integration, the estimator-aware feasibility test and the measured trade-offs, not a demonstrated safety improvement from health-aware deconfliction.

Why the traffic-level effect is small here. No arm ever depleted a battery, because onboard feasibility checks with a 20% reserve already kept every drone landable; the traffic manager can only add value when drones compete for scarce landing capacity or when conflict manoeuvres consume a significant share of the margin. Pad occupancy is not modelled as a resource that can be over-subscribed, and the energy overhead of conflict resolution on a 2-km campus route is small. Scenarios with longer routes, limited pad capacity and denser low-margin traffic are needed to test whether the arbitration and the weights matter.

Where to hold. The seed-13 forced landing shows that the hold-short point matters. Arrivals hold at the edge of the destination's tube on their current layer, which can be the layer the critical drone approaches on (in seed 13 it was, since ADMM only couples drones on the same layer), so the two can meet near the pad, and the resulting avoidance manoeuvre draws current exactly when a failed cell has the least headroom. Holding on a different layer, or outside the approach path, and limiting the avoidance demand on a drone whose weakest-cell margin is small are straightforward refinements; they were not evaluated here.

Thermal response. The heat-wave result shows that a threshold-and-hold thermal response is too slow at high ambient temperature. A predictive response that diverts when the predicted time to the FAULT temperature falls below the time to the nearest pad, for example as an STL specification $\mathbf{G}(T_p \leq T_{\max})$ with robustness-driven intervention, is the natural next step.

Threats to validity. (i) The battery, power and thermal models are documented but not calibrated against a specific cell or airframe; public flight-energy data for delivery multicopters [33] would allow this. (ii) Sensors and faults are simulated, and the BMS is assumed to identify cell offsets and R_0 online. (iii) One map, one horizon and 20 seeds per cell were used; effects smaller than the reported intervals cannot be excluded. (iv) The fleet includes 450-kg passenger eVTOLs in 20-m corridors next to 6-kg drones, which is not a realistic campus mix. (v) Wind changes airspeed and power but does not displace the drones. (vi) Health telemetry is trusted; a drone that falsely reports CRITICAL would obtain right of way, and protecting health reports in the way W-MSR protects positions is open. (vii) ADMM is used as a mitigation layer with a convexified, per-tick separation constraint and provides no formal guarantee. (viii) Earlier experiments with the project's first, three-corridor engine are not reported, because that engine evaluated separation horizontally and its results were superseded.

10 Conclusion

This paper connected the battery management system of each drone to the traffic manager of a drone fleet. A 1-RC pack model with weakest-cell energy, an EKF on state of charge, polarisation voltage and capacity, and a chance-constrained feasibility test with a voltage-sag check feed every onboard and fleet-level decision. The traffic controller uses the same state for right of way, endurance-ordered pad arbitration and bounded energy-margin weights in consensus ADMM. In 460 paired-seed runs on a campus corridor network, the estimator was calibrated and never optimistic in nominal flight, and the chance-constrained check traded in-flight diversions for pre-flight holds without losing deliveries; health-aware arbitration and weighting brought no measurable benefit in the tested scenarios and, in one run, a hold-short placement that led to a forced landing, and a threshold-based thermal response proved too slow at 44 °C. Future work will calibrate the models on flight data, add predictive thermal and energy specifications to the STL monitor, protect health telemetry against falsification, and test the fleet-level mechanisms in scenarios where landing capacity and energy margins are genuinely scarce.

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